

spark

<pyspark.sql.connect.session.SparkSession at 0xffbe64115430>

```
from pyspark.sql import SparkSession
```

```
spark=SparkSession.builder.appName('superstore_sales').getOrCreate()
```

```
df = spark.table("workspace.default.superstore_sales")
```

```
df.na.drop(how="any").show()
```

```
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+
-+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+
|      order_id|order_date| ship_date|      ship_mode|      customer_name|
segment|          state|          country|market|          region|
product_id|          category|sub_category|
product_name|sales|quantity|discount|
profit|shipping_cost|order_priority|year|
+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+
-+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+
|   AG-2011-2040|2011-01-01|2011-01-06|Standard Class|   Toby Braunhardt|
Consumer|      Constantine|      Algeria|Africa|      Africa|OFF-TEN-
10000025|Office Supplies|      Storage|Tenex Lockers, Blue|   408|      2|
0.0|   106.14|      35.46|      Medium|2011|
|   IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer| New South Wales|      Australia| APAC|      Oceania| OFF-SU-
10000618|Office Supplies|      Supplies|Acme Trimmer, Hig...|   120|      3|
0.1|   36.036|      9.72|      Medium|2011|
|   HU-2011-1220|2011-01-01|2011-01-05| Second Class|      Annie Thurman|
Consumer|      Budapest|      Hungary| EMEA|      EMEA|OFF-TEN-
10001585|Office Supplies|      Storage|Tenex Box, Single...|   66|      4|
0.0|   29.64|      8.17|      High|2011|
|IT-2011-3647632|2011-01-01|2011-01-05| Second Class|      Eugene Moren|Home
Office|      Stockholm|      Sweden| EU|      North| OFF-PA-
10001492|Office Supplies|      Paper|Enermax Note Card...|   45|      3|
0.5|  -26.055|      4.82|      High|2011|
|   IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer| New South Wales|      Australia| APAC|      Oceania| FUR-FU-
10003447|      Furniture|Furnishings|Eldon Light Bulb,...|   114|      5|
0.1|   37.77|      4.7|      Medium|2011|
|   IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer| New South Wales|      Australia| APAC|      Oceania| OFF-PA-
10001968|Office Supplies|      Paper|Eaton Computer Pr...|   55|      2|
0.1|   15.342|      1.8|      Medium|2011|
|   CA-2011-1510|2011-01-02|2011-01-06|Standard Class|      Magdelene Morse|
Consumer|      Ontario|      Canada|Canada|      Canada|TEC-OKI-
10002750|      Technology|      Machines|Okidata Inkjet, W...|   314|      1|
```

0.0	3.12	24.1	Medium	2011				
IN-2011-79397	2011-01-03	2011-01-03	Same Day	Kean Nguyen				
Corporate	New South Wales	Australia	APAC	Oceania	OFF-AP-			
10000304	Office Supplies	Appliances	Hoover Microwave, ...	276	1			
0.1	110.412	125.32	Critical	2011				
ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale				
Consumer	Auckland	New Zealand	APAC	Oceania	TEC-CO-			
10004182	Technology	Copiers	Hewlett Wireless ...	912	4			
0.4	-319.464	107.1	Low	2011				
IZ-2011-4680	2011-01-03	2011-01-07	Standard Class	Lindsay Williams				
Corporate	Ninawa	Iraq	EMEA	EMEA	FUR-NOV-			
10002791	Furniture	Chairs	Novimex Swivel St...	667	4			
0.0	253.32	81.26	High	2011				
IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks				
Consumer	National Capital	Philippines	APAC	Southeast Asia	OFF-ST-			
10003020	Office Supplies	Storage	Tenex Lockers, In...	338	3			
0.45	-122.8005	33.75	High	2011				
IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks				
Consumer	National Capital	Philippines	APAC	Southeast Asia	FUR-TA-			
10002797	Furniture	Tables	Chromcraft Round ...	211	1			
0.55	-70.3995	21.32	High	2011				
ES-2011-4869686	2011-01-03	2011-01-07	Standard Class	Dorothy Dickinson				
Consumer	England	United Kingdom	EU	North	FUR-BO-			
10000728	Furniture	Bookcases	Dania Corner Shel...	854	7			
0.0	290.43	12.56	Medium	2011				
IN-2011-33652	2011-01-03	2011-01-09	Standard Class	Dennis Pardue	Home			
Office	Sarawak	Malaysia	APAC	Southeast Asia	TEC-CO-			
10000594	Technology	Copiers	Hewlett Fax and C...	193	1			
0.0	50.13	10.4	Medium	2011				
ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale				
Consumer	Auckland	New Zealand	APAC	Oceania	FUR-CH-			
10000214	Furniture	Chairs	Hon Rocking Chair...	159	2			
0.4	-95.676	10.07	Low	2011				
MX-2011-160234	2011-01-03	2011-01-07	Standard Class	Stewart Visinsky				
Consumer	Guatemala	Guatemala	LATAM	Central	TEC-PH-			
10002647	Technology	Phones	Nokia Headset, VoIP	195	4			
0.0	44.88	8.43	Medium	2011				
IR-2011-770	2011-01-03	2011-01-07	Standard Class	Jas O'Carroll				
Consumer	Yazd	Iran	EMEA	EMEA	OFF-BRE-			
10003081	Office Supplies	Appliances	Breville Coffee G...	123	2			
0.0	42.9	8.41	Medium	2011				
ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale				
Consumer	Auckland	New Zealand	APAC	Oceania	TEC-AC-			
10002881	Technology	Accessories	Belkin Numeric Ke...	69	2			
0.4	3.42	8.34	Low	2011				
ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale				
Consumer	Auckland	New Zealand	APAC	Oceania	FUR-CH-			
10000666	Furniture	Chairs	SAFCO Chairmat, B...	69	2			
0.4	-26.412	8.17	Low	2011				
ID-2011-12596	2011-01-03	2011-01-08	Standard Class	Chris McAfee				
Consumer	Nakhon Ratchasima	Thailand	APAC	Southeast Asia	OFF-ST-			
10002066	Office Supplies	Storage	Smead File Cart, ...	135	2			
0.47	-45.9018	7.74	Medium	2011				

```

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+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+

```

only showing top 20 rows

```
df.columns
```

```

['order_id',
 'order_date',
 'ship_date',
 'ship_mode',
 'customer_name',
 'segment',
 'state',
 'country',
 'market',
 'region',
 'product_id',
 'category',
 'sub_category',
 'product_name',
 'sales',
 'quantity',
 'discount',
 'profit',
 'shipping_cost',
 'order_priority',
 'year']

```

Discribe Summary of Data

```
display(df.describe())
```

summary	order_id	ship_mode	customer_name	segment	state	country
count	51290	51290	51290	51290	51290	51290
mean	null	null	null	null	null	null
stddev	null	null	null	null	null	null
min	AE-2011-9160	First Class	Aaron Bergman	Consumer	'Ajman	Afghanistan
max	ZI-2014-9650	Standard Class	Zuschuss Donatelli	Home Office	Zürich	Zimbabwe

```
display(df.summary())
```

summary	order_id	ship_mode	customer_name	segment	state	country
count	51290	51290	51290	51290	51290	51290
mean	null	null	null	null	null	null
stddev	null	null	null	null	null	null
min	AE-2011-9160	First Class	Aaron Bergman	Consumer	'Ajman	Afghanistan
25%	null	null	null	null	null	null
50%	null	null	null	null	null	null

Show Null Value And Removing

```
# Check null values per column (as percentage)
from pyspark.sql.functions import col, count, when

total_rows = df.count()
null_counts = df.select([count(when(col(c).isNull(), c)).alias(c) for c in
                        df.columns])
null_percentages = null_counts.select([(col(c) / total_rows * 100).alias(c)
                                       for c in df.columns])

print("Null value percentages by column:")
display(null_percentages)

# Drop rows with any null values
df_cleaned = df.na.drop(how="any")
print(f"\nOriginal row count: {total_rows}")
print(f"Cleaned row count: {df_cleaned.count()}")
```

Null value percentages by column:

order_id	order_date	ship_date	ship_mode	customer_name	segment	state	country
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Original row count: 51290

Cleaned row count: 51290

```
df.printSchema
```

Column Filter

```
df.show(truncate=True)
```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      order_id|order_date| ship_date|      ship_mode|      customer_name|
segment|      state|      country|market|      region|
product_id|      category|sub_category|
product_name|sales|quantity|discount|
profit|shipping_cost|order_priority|year|
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+
|      AG-2011-2040|2011-01-01|2011-01-06|Standard Class|      Toby Braunhardt|
Consumer|      Constantine|      Algeria|Africa|      Africa|OFF-TEN-
10000025|Office Supplies|      Storage| Tenex Lockers, Blue|      408|      2|
0.0|      106.14|      35.46|      Medium|2011|
|      IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer|      New South Wales|      Australia| APAC|      Oceania| OFF-SU-
10000618|Office Supplies|      Supplies|Acme Trimmer, Hig...|      120|      3|
0.1|      36.036|      9.72|      Medium|2011|
|      HU-2011-1220|2011-01-01|2011-01-05| Second Class|      Annie Thurman|
Consumer|      Budapest|      Hungary| EMEA|      EMEA|OFF-TEN-
10001585|Office Supplies|      Storage|Tenex Box, Single...|      66|      4|
0.0|      29.64|      8.17|      High|2011|
|IT-2011-3647632|2011-01-01|2011-01-05| Second Class|      Eugene Moren|Home
Office|      Stockholm|      Sweden| EU|      North| OFF-PA-
10001492|Office Supplies|      Paper|Enermax Note Card...|      45|      3|
0.5|      -26.055|      4.82|      High|2011|
|      IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer|      New South Wales|      Australia| APAC|      Oceania| FUR-FU-
10003447|      Furniture|Furnishings|Eldon Light Bulb,...|      114|      5|
0.1|      37.77|      4.7|      Medium|2011|
|      IN-2011-47883|2011-01-01|2011-01-08|Standard Class|      Joseph Holt|
Consumer|      New South Wales|      Australia| APAC|      Oceania| OFF-PA-
10001968|Office Supplies|      Paper|Eaton Computer Pr...|      55|      2|
0.1|      15.342|      1.8|      Medium|2011|
|      CA-2011-1510|2011-01-02|2011-01-06|Standard Class|      Magdelene Morse|
Consumer|      Ontario|      Canada|Canada|      Canada|TEC-OKI-
10002750|      Technology|      Machines|Okidata Inkjet, W...|      314|      1|
0.0|      3.12|      24.1|      Medium|2011|
|      IN-2011-79397|2011-01-03|2011-01-03|      Same Day|      Kean Nguyen|
Corporate|      New South Wales|      Australia| APAC|      Oceania| OFF-AP-
10000304|Office Supplies|      Appliances|Hoover Microwave,...|      276|      1|
0.1|      110.412|      125.32|      Critical|2011|
|      ID-2011-80230|2011-01-03|2011-01-09|Standard Class|      Ken Lonsdale|
Consumer|      Auckland|      New Zealand| APAC|      Oceania| TEC-CO-
10004182|      Technology|      Copiers|Hewlett Wireless ...|      912|      4|
0.4|      -319.464|      107.1|      Low|2011|
|      IZ-2011-4680|2011-01-03|2011-01-07|Standard Class|      Lindsay Williams|
```

Corporate	Ninawa	Iraq	EMEA	EMEA	FUR-NOV-10002791	
	Furniture	Chairs	Novimex Swivel St...		667	4
0.0	253.32	81.26	High	2011		
	IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks	
Consumer	National Capital	Philippines	APAC	Southeast Asia	OFF-ST-10003020	
	Office Supplies	Storage	Tenex Lockers, In...		338	3
0.45	-122.8005	33.75	High	2011		
	IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks	
Consumer	National Capital	Philippines	APAC	Southeast Asia	FUR-TA-10002797	
	Furniture	Tables	Chromcraft Round ...		211	1
0.55	-70.3995	21.32	High	2011		
	ES-2011-4869686	2011-01-03	2011-01-07	Standard Class	Dorothy Dickinson	
Consumer	England	United Kingdom	EU	North	FUR-BO-10000728	
	Furniture	Bookcases	Dania Corner Shel...		854	7
0.0	290.43	12.56	Medium	2011		
	IN-2011-33652	2011-01-03	2011-01-09	Standard Class	Dennis Pardue	Home
Office	Sarawak	Malaysia	APAC	Southeast Asia	TEC-CO-10000594	
	Technology	Copiers	Hewlett Fax and C...		193	1
0.0	50.13	10.4	Medium	2011		
	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	
Consumer	Auckland	New Zealand	APAC	Oceania	FUR-CH-10000214	
	Furniture	Chairs	Hon Rocking Chair...		159	2
0.4	-95.676	10.07	Low	2011		
	MX-2011-160234	2011-01-03	2011-01-07	Standard Class	Stewart Visinsky	
Consumer	Guatemala	Guatemala	LATAM	Central	TEC-PH-10002647	
	Technology	Phones	Nokia Headset, VoIP		195	4
0.0	44.88	8.43	Medium	2011		
	IR-2011-770	2011-01-03	2011-01-07	Standard Class	Jas O'Carroll	
Consumer	Yazd	Iran	EMEA	EMEA	OFF-BRE-10003081	
	Office Supplies	Appliances	Breville Coffee G...		123	2
0.0	42.9	8.41	Medium	2011		
	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	
Consumer	Auckland	New Zealand	APAC	Oceania	TEC-AC-10002881	
	Technology	Accessories	Belkin Numeric Ke...		69	2
0.4	3.42	8.34	Low	2011		
	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	
Consumer	Auckland	New Zealand	APAC	Oceania	FUR-CH-10000666	
	Furniture	Chairs	SAFCO Chairmat, B...		69	2
0.4	-26.412	8.17	Low	2011		
	ID-2011-12596	2011-01-03	2011-01-08	Standard Class	Chris McAfee	
Consumer	Nakhon Ratchasima	Thailand	APAC	Southeast Asia	OFF-ST-10002066	
	Office Supplies	Storage	Smead File Cart, ...		135	2
0.47	-45.9018	7.74	Medium	2011		

```

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+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+

```

only showing top 20 rows

Row Filtering

Filter sales data between 2011 and 2012

```
sales_2011_2012 = df.filter((df.order_date >= "2011-01-01") & (df.order_date
    <= "2012-12-31"))
display(sales_2011_2012)
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest
IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales

```
from pyspark.sql.functions import col
```

Filter sales data between 2011 and 2012 using col() and between()

```
sales_2011_2012 = df.filter(col("order_date").between("2011-01-01", "2012-12-31"))
display(sales_2011_2012)
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest
IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales

Example: Select specific columns using col()

```
df.select(col("order_id"), col("customer_name"), col("sales")).show()
```

```

+-----+-----+-----+
| order_id | customer_name | sales |
+-----+-----+-----+
| AG-2011-2040 | Toby Braunhardt | 408 |
| IN-2011-47883 | Joseph Holt | 120 |
| HU-2011-1220 | Annie Thurman | 66 |
| IT-2011-3647632 | Eugene Moren | 45 |
| IN-2011-47883 | Joseph Holt | 114 |
| IN-2011-47883 | Joseph Holt | 55 |
| CA-2011-1510 | Magdelene Morse | 314 |
| IN-2011-79397 | Kean Nguyen | 276 |
| ID-2011-80230 | Ken Lonsdale | 912 |
| IZ-2011-4680 | Lindsay Williams | 667 |
| IN-2011-65159 | Larry Blacks | 338 |
| IN-2011-65159 | Larry Blacks | 211 |
| ES-2011-4869686 | Dorothy Dickinson | 854 |
| IN-2011-33652 | Dennis Pardue | 193 |
| ID-2011-80230 | Ken Lonsdale | 159 |
| MX-2011-160234 | Stewart Visinsky | 195 |
| IR-2011-770 | Jas O'Carroll | 123 |
| ID-2011-80230 | Ken Lonsdale | 69 |
| ID-2011-80230 | Ken Lonsdale | 69 |
| ID-2011-12596 | Chris McAfee | 135 |
+-----+-----+-----+

```

only showing top 20 rows

```

data= df.select("order_id","customer_name","sales","profit")
display(data)

```

order_id	customer_name	sales	profit
AG-2011-2040	Toby Braunhardt	408	106.14
IN-2011-47883	Joseph Holt	120	36.036
HU-2011-1220	Annie Thurman	66	29.64
IT-2011-3647632	Eugene Moren	45	-26.055
IN-2011-47883	Joseph Holt	114	37.77
IN-2011-47883	Joseph Holt	55	15.342
CA-2011-1510	Magdelene Morse	314	3.12
IN-2011-79397	Kean Nguyen	276	110.412
ID-2011-80230	Ken Lonsdale	912	-319.464

```

sales_2011_2012 = df.filter((df.order_date >= "2011-01-01") & (df.order_date
    <= "2012-12-31")).select("order_id", "customer_name", "sales",
    "profit", "order_date")
display(sales_2011_2012)

```

order_id	customer_name	sales	profit	order_date
AG-2011-2040	Toby Braunhardt	408	106.14	2011-01-01
IN-2011-47883	Joseph Holt	120	36.036	2011-01-01
HU-2011-1220	Annie Thurman	66	29.64	2011-01-01
IT-2011-3647632	Eugene Moren	45	-26.055	2011-01-01
IN-2011-47883	Joseph Holt	114	37.77	2011-01-01
IN-2011-47883	Joseph Holt	55	15.342	2011-01-01
CA-2011-1510	Magdelene Morse	314	3.12	2011-01-02
IN-2011-79397	Kean Nguyen	276	110.412	2011-01-03
ID-2011-80230	Ken Lonsdale	912	-319.464	2011-01-03

```
display(df.select("category").distinct())
```

category
Office Supplies
Furniture
Technology

```
# Filter only Office Supplies category
office_supplies = df.filter(df.category == "Office Supplies")
display(office_supplies)
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest
IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm
IN-2011-	2011-01-01	2011-01-	Standard			New South

```
from pyspark.sql.functions import regexp_replace, col
```

```
df_with_revenu = df.withColumn("revenu", (regexp_replace(col("sales"), ",",
    "").cast("double") + col("quantity")) - col("discount"))
display(df_with_revenu.select("order_id", "customer_name", "sales",
    "quantity", "discount", "revenu"))
```

order_id	customer_name	sales	quantity	discount	revenu
AG-2011-2040	Toby Braunhardt	408	2	0.0	410.0
IN-2011-47883	Joseph Holt	120	3	0.1	122.9
HU-2011-1220	Annie Thurman	66	4	0.0	70.0
IT-2011-3647632	Eugene Moren	45	3	0.5	47.5
IN-2011-47883	Joseph Holt	114	5	0.1	118.9
IN-2011-					

2) How many totoal profit generate By category and sub_category

```
df.select('category').distinct().collect()
```

```
[Row(category='Office Supplies'),
 Row(category='Furniture'),
 Row(category='Technology')]
```

```
df.groupBy("category").agg({'profit': 'sum'}).show()
```

```
+-----+-----+
|      category|      sum(profit)|
+-----+-----+
|Office Supplies|518473.8342999985|
|      Furniture|286782.2538000002|
|      Technology|663778.7331799994|
+-----+-----+
```

```
prof_bysubcate=df.groupBy("sub_category").agg({'profit': 'sum'})
display(prof_bysubcate)
```

sub_category	sum(profit)
Storage	108461.48979999997
Supplies	22583.263099999986
Paper	59207.682699999874
Furnishings	46967.425499999925
Machines	58867.87299999998
Appliances	141680.58940000017
Copiers	258567.5481799999
Chairs	141973.79749999967
Tables	-64083.38869999998

```
prof_bysubcate.orderBy(['sum(profit)'], ascending = [False]).show(10)
```

```
+-----+-----+
|sub_category|      sum(profit)|
+-----+-----+
|    Copiers| 258567.5481799999|
|    Phones|    216717.0058|
| Bookcases|161924.41950000016|
|    Chairs|141973.79749999967|
| Appliances|141680.58940000017|
| Accessories|129626.30619999964|
|    Storage|108461.48979999997|
|    Binders| 72449.84600000024|
|    Paper|59207.682699999874|
|    Machines| 58867.87299999998|
+-----+-----+
only showing top 10 rows
```

filter function

```
# Using equals condition
```

```
united_state= df.filter(df.country == "United States")
display(united_state)
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
CA-2011-103800	2011-01-04	2011-01-08	Standard Class	Darren Powers	Consumer	Texas
CA-2011-112326	2011-01-05	2011-01-09	Standard Class	Phillina Ober	Home Office	Illinois

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
CA-2011-112326	2011-01-05	2011-01-09	Standard Class	Phillina Ober	Home Office	Illinois
CA-		2011-01-	Standard		Home	

```
df_filter= df.filter((df['order_date'] >= '2011-01-01') & (df['order_date']
<= '2012-01-07'))
display(df_filter)
# Using SQL condition`
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest
IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales

```
data=df.filter(df["order_date"].between('2011-01-01', '2012-01-07'))
display(data)
```

order_id	order_date	ship_date	ship_mode	customer_name	segment	state
AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	Budapest
IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	Stockholm
IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	New South Wales